Neural Solvers for Data-Driven Combinatorial Optimization with Guarantees

Nikolaos Karalias

I develop mathematical tools that enable gradient-based methods to solve challenging combinatorial optimization problems and carry out discrete computations. My work centers on designing continuous formulations that allow these methods to efficiently explore large combinatorial spaces while exploiting the underlying problem structure. I study how the choice of formulation affects the computations an algorithm can perform, the solutions it can reach, and the guarantees we can establish.

Neural networks are a central focus of this research, and I have a particular interest in scalable methods based on self-supervised training. One of my long-term goals is to build general-purpose neural solvers that combine the ability to adapt to data with the feasibility and approximation guarantees of classical discrete algorithms.

1 Introduction

Combinatorial and discrete optimization problems arise in routing, resource allocation, and the design of complex systems. A wide range of algorithms can be used to solve them. At one end of the spectrum, learning-based methods can recognize and exploit structure shared across problem instances, but their ability to adapt to data may come at the cost of reliability. At the other end, classical algorithms often provide guarantees on feasibility and solution quality, but may leave problem-specific structure unexploited and miss opportunities to obtain better solutions or reach them more efficiently. A central goal of my research is to strike the best possible balance between these two ends of the spectrum, developing methods that take advantage of problem structure while retaining guarantees and keeping the cost of learning and computation manageable.

Research thesis. Design continuous representations and learning algorithms that turn the structure of discrete problems into efficient computation and explicit guarantees.

Existing approaches offer different ways to balance flexibility, computational cost, and guarantees. Learning can guide established optimization algorithms, while end-to-end neural methods can search for solutions directly. My approach develops self-supervised objectives and algorithmic components that connect neural predictions to discrete solutions and their properties. By learning from the objective and constraints themselves, I aim to reduce dependence on expensive solved examples while identifying the assumptions under which the resulting solver can be trusted.

2 Foundations of my research program

Throughout my PhD and postdoctoral work, I have studied how the mathematical representation of a discrete problem shapes both the learning process and the properties of the resulting solutions. I introduced a probabilistic approach to self-supervised combinatorial optimization, developed neural set function extensions, and led work connecting message passing with approximation algorithms. Subsequent work with my collaborators extended these ideas to problems with discrete constraints. These contributions provide the foundation for the research program I describe here, in which I will study how representation, optimization, and learning can work together to make neural solvers more efficient and reliable.

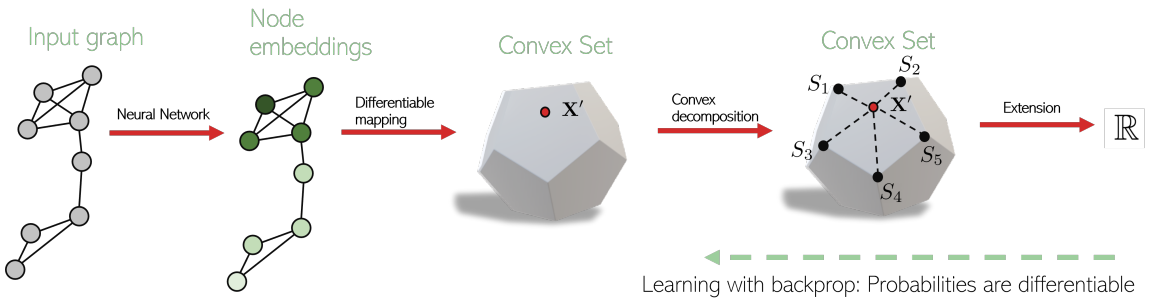


Figure 1: Learning in the extensions framework.

Principled and scalable learning frameworks for neural combinatorial optimization. An important consideration when training machine learning models is the availability of labeled data. Training with standard supervised approaches in combinatorial optimization would require access to large amounts of solved instances. For difficult problems, obtaining solutions is computationally expensive, which highlights the need for an efficient and scalable learning paradigm.

In "Erdős Goes Neural" [Karalias and Loukas, 2020] (NeurIPS 2020 **Oral**), I described the first principled approach to solving combinatorial optimization problems with neural networks without access to any labels. The key idea was that, under suitable assumptions, one could construct differentiable loss functions that encode the objective value and constraint violation of solutions to the problem. These loss functions lead to solutions of certified quality and were also coupled to a principled rounding scheme that was guaranteed to preserve solution quality after discretization. This paper inspired impactful follow-up work, including *breakthroughs in the Hadwiger-Nelson problem using neural networks* [Mundinger et al., 2025], as well as applications to large-scale instances and other optimization problems (e.g., [Qiu et al., 2022, Li et al., 2023, 2024b, Geng et al., 2025]).

To further generalize and expand the toolkit for unsupervised neural combinatorial optimization, I proposed a framework for neural set function extensions [Karalias et al., 2022](NeurIPS 2022). The Erdős framework relied on closed-form expectations of objective and constraint functions over products of independent marginals to train neural networks. In contrast, the extensions framework dispenses with this distributional requirement and instead constructs a continuous, differentiable extension of the discrete objective. The output of a neural network is converted to a distribution through an appropriate differentiable map, and then that distribution is used to deterministically and exactly calculate the expected value of the objective of the problem on the inputs. Figure 1 provides a high-level intuitive overview of the pipeline. Importantly, this approach can be implemented with only black-box access to the objective function, enabling a much broader applicability for the method. Extensions have since been used to achieve state-of-the-art results for the quadratic assignment problem [Nerem et al., 2024] and for other applications such as graph neural network interpretability [Chen et al., 2024]. Alongside these frameworks for discrete objectives, I also study how learning can interact with discrete algorithms. In our work on gradient descent programs, my collaborators and I show how gradient descent can express discrete computation [Karalias et al., 2026b]. We establish optimization guarantees for certain semidefinite formulations and develop a variational approach to backward passes through discrete algorithms, including richer representations based on semidefinite relaxations. The broader aim is to learn from the outcomes of discrete computations while retaining the algorithmic structure that makes their behavior reliable.

Approximation capabilities of neural solvers. Unlike classical solvers and algorithms with theoretical guarantees, neural networks largely remain a black-box. Furthermore, they are generally tailored to distributional or *average-case* problem solving. Therefore, securing guarantees for their worst-case performance in combinatorial problems is a rather challenging task. To make matters more difficult, graph neural networks, which were some of the most prominent models in the literature for neural CO, were

thought to be fairly limited in terms of approximation power [Sato et al., 2019, Gamarnik, 2023].

In the paper "Are Graph Neural Networks Optimal Approximation Algorithms?" (NeurIPS 2024 **Spotlight**) [Yau et al., 2024], I led a collaboration with students from MIT and Harvard that demonstrated that graph neural networks can be used to implement polynomial-time approximation algorithms with the *best known approximation guarantees for all constraint satisfaction problems (CSPs).* The main idea behind this approach is to show that graph neural networks can efficiently implement an SDP solver for the Sum-of-Squares relaxation of CSPs. Furthermore, assuming the Unique Games Conjecture, the achieved approximation ratios are optimal [Raghavendra, 2008]. These theoretical results solidified the potential that graph neural network architectures have for solving hard combinatorial problems and also led to practical GNN architectures that can be trained without supervision to achieve strong results on multiple CSPs. The ability of a neural architecture to represent a strong algorithm also raises a complementary question about what gradient-based optimization can actually find. We investigate this question for SAT in "On the Fundamentals of Gradient-Based SAT Solvers" (NeurIPS 2026) [Karalias et al., 2026c]. My collaborators and I establish sufficient conditions for convergence toward satisfying solutions and identify structural limitations of both the optimization dynamics and the neural architectures we analyze. We also show how transformations that preserve satisfiability can change the difficulty of the optimization problem and overcome certain limitations.

Producing feasible solutions for combinatorial problems. A persistent challenge for the field has been to train neural networks that can reliably generate feasible solutions for discrete optimization problems [Smith, 1999, Bu et al., 2024]. Since the space of possible constraints is vast, it is rather difficult to devise a general all-encompassing approach. To this end, I proposed a general framework for extensions [Karalias et al., 2025] (NeurIPS 2025) that tackles precisely this challenge. It offers a general optimization framework for combinatorial optimization with constraints that covers a large class of problems. More specifically, for any problem whose feasible set is a convex polytope that admits an efficient linear optimization oracle, we use a constructive version of the Carathéodory theorem to create distributions of feasible solutions that allow exact calculation of the expected value of the objective. Many NP-hard combinatorial problems fall under this umbrella (Travelling Salesperson, Maximum Coverage, Constrained Partitioning, etc.). This approach leads to the best known results for a neural baseline on cardinality-constrained submodular maximization (i.e., coverage and facility location problems).

Recent work on Frank–Wolfe methods and parallel algorithms extends this framework toward more flexible and scalable neural solvers [Rafiey et al., 2026, Soleymani et al., 2026]. We develop ways to convert neural predictions into feasible solutions across problem families with suitable optimization oracles, and make these constructions more efficient through parallel computation. Together, these projects broaden the range of models and problems that can benefit from explicit constraint handling. The same concern with feasibility arises when learned models generate structured outputs, where enforcing constraints can also change the distribution of the samples. My work on constrained diffusion models examines this relationship and identifies conditions under which the dependencies among constraints support reliable generation [Karalias et al., 2026a]. These results motivate a broader direction in which I will investigate how sparse dependencies can make constrained generation more effective. I plan to extend these insights to settings such as language generation, where satisfying requirements must also preserve the usefulness and coherence of the generated output.

3 Toward efficient and general neural solvers with guarantees

The goal of the research program is to build a comprehensive toolkit that can be used to tackle mathematical and combinatorial optimization problems in their full generality. I will provide an overview of the main pillars of my future research agenda. For the purposes of clarity and organization, I will discuss each one separately. However, these pillars share several important common threads, and interesting research questions may also arise at their boundaries.

Self-supervised learning with differentiable proxies as an alternative to reinforcement learning. Reinforcement learning has been a leading problem-solving paradigm for combinatorial optimization primarily due to its versatility and wide applicability. One of its drawbacks is that stochastic gradient estimation can be challenging and computationally inefficient in structured high-dimensional combinatorial spaces since it requires efficient sampling, has high variance, and a large number of calls to the reward function [Grathwohl et al., 2018, Schulman et al., 2015]. My research program advocates for the following shift in perspective. Instead of relying on sampling and stochastic estimation, the *focus of my research will be on constructing differentiable proxies that can be evaluated deterministically, efficiently, and that lead to data-efficient self-supervised learning.* The framework of extensions that I have described provides some first steps towards this goal by enabling exact computation of expected values of objectives over distributions of feasible solutions [Karalias et al., 2025] in a differentiable manner. Since there is explicit control in how extensions are constructed, certain desirable properties (e.g., smoothness) can be built into the proxies that can lead to convergence guarantees. The overarching goal is to enable efficient data-driven optimization in high-dimensional, non-convex settings where the objective and constraints are black boxes. This direction has both practical and theoretical dimensions. On the theoretical front, it will require developing a deeper mathematical understanding of non-convex optimization with such smooth proxies using gradient descent. Additionally, a key design component of those proxies will be their dimensionality. Both classical approximation algorithms and neural networks suggest that higher-dimensional representations can offer favorable optimization properties and support more powerful algorithms. Pursuing this direction will require a deeper understanding of the relationship between discrete structure and continuous optimization, drawing on ideas from combinatorial and algebraic geometry.

On the practical front, efficient and parallelizable implementations of those ideas will be necessary to make those tools scalable to large instances. Therefore, the challenges posed by this direction present an excellent opportunity for broad collaborations with researchers in various fields of mathematics and engineering. The need for efficiency also leads us to the next direction of this research program.

General representations and data efficiency via augmentations and polynomial-time reductions. Reductions provide a principled starting point for studying what neural solvers can share across different combinatorial problems. The relationships established by complexity theory suggest opportunities for common representations, but useful transfer also depends on what a reduction preserves about feasible solutions and their quality. I will investigate how these relationships can support learning across problems and reduce the amount of data needed to train effective solvers. My aim is to understand when formal connections between problems translate into shared algorithmic knowledge, and how that knowledge can help a model adapt beyond the setting in which it was trained.

Another important aspect of this direction relates to data augmentations. Augmentations can play an important role in classical solvers [Akiba and Iwata, 2016, Davis and Putnam, 1960], but their effect on neural CO has not been thoroughly explored. Prior work has shown that certain instance augmentations can dramatically reduce the number of labeled instances required to train neural SAT solvers[Duan et al., 2022]. My work on label-preserving augmentations and the foundations of gradient-based SAT solving provides further evidence that the representation of an instance can influence both learning and direct optimization [Kostin et al., 2025, Karalias et al., 2026c]. Crucially, data augmentations can be beneficial for gradient-based optimization even in the absence of a neural network, i.e., when minimizing a non-convex objective directly with gradient descent. This implies that progress in this direction may not only be beneficial for neural solvers but may lead to new insights into aspects of non-convex optimization and classical combinatorial optimization.

Securing optimality certificates and approximation guarantees in practice. The connections between neural network architectures and convex optimization solvers that have been highlighted in my work [Yau et al., 2024], and elsewhere in the literature [Qian et al., 2023, Li et al., 2024a, 2025, Chen et al., 2022], highlight the benefits of neural networks that can emulate algorithms. Combined with recent

advances in certified non-convex optimization [Beugnot et al., 2023, Wang, 2022], it is clear that these ideas point to an implementation paradigm in which neural solvers *leverage high-dimensional representations to generate solutions with certificates and worst-case approximation guarantees*. Furthermore, I plan to use those certificates to help neural solvers improve upon classical algorithms. The key challenge in this direction is guaranteeing that the outputs of a neural network will precisely match the outputs of an algorithm that has a performance guarantee. By achieving that, the neural network will inherit the performance guarantee while also being able to improve upon its solutions in a data-driven fashion.

Understanding the contribution of learning. An important question throughout this program is when learning across instances improves on solving each instance directly. Some of the mathematical tools I develop can benefit both approaches, so I want to understand which gains come from the formulation itself and which depend on what the model has learned. I will study this distinction with attention to the quality of the resulting solutions, the computation required to obtain them, and how well the benefits extend to new settings. This perspective will help identify where neural solvers offer a lasting advantage and where progress in the underlying optimization methods is more valuable.

Real-world applications, competition benchmarks, and improving LLMs. Chip placement will provide an engineering application alongside SAT. I will investigate whether the optimization objectives and constraint-handling methods developed in this program can improve placement quality or reduce the computational effort needed to find good solutions. Comparisons with classical optimization and reinforcement-learning approaches will help establish where the methods contribute in practice [Mirhoseini et al., 2020]. This application will also test how well the mathematical ideas carry over to the objectives, constraints, and scale of an engineering problem.

Synthetic instances and SAT competitions [Ganesh and Vardi, 2020] will be another important element of benchmarking for my methods. Those kinds of instances provide an excellent sandbox for prototyping and controlled experimentation when developing new methods. This leads me to a major long-term benchmarking goal for the project which will be to *create the first neural solver that can outperform state-of-the-art classical solvers in a SAT competition*. Currently, this appears to be out of reach due to the exceptional computational efficiency of SAT solvers. This kind of benchmarking goal relies on competitions with a decades-long track record and avoids many of the evaluation pitfalls that other machine learning models such as Graph Neural Networks [Bechler-Speicher et al., 2025] and LLMs [Thaker et al., 2025] have historically faced. For example, it is highly unlikely that training data for real-world combinatorial problems will be inadvertently contaminated with exact solutions, which makes these domains particularly suitable for rigorous evaluation of learning-based solvers.

I also study compositional reasoning in language models through a combinatorial lens, focusing on how models allocate a limited token budget across the tasks that a larger problem brings together. Our recent work on tasks that compete for tokens provides a starting point for this investigation. We examine how performance changes when independent tasks share a prompt, and find that task grouping can reduce token use while preserving measured accuracy in the settings evaluated [Li et al., 2026]. Building on these observations, I will investigate how the structure and difficulty of individual tasks influence budget allocation and whether better allocation can help models reason more effectively when those tasks are combined. Combinatorial problems provide a useful setting for these questions because their objectives and constraints offer clear ways to assess the quality of an answer. My aim is to improve the efficiency and reliability of compositional reasoning while understanding how well those improvements extend to unfamiliar combinations of tasks.

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